

Advanced Time Series Problem Set

1) Given an ACF with slow decay and PACF cutoff at lag 2, identify model and justify. 2) A monthly series shows spikes at lags 12 and 24 in ACF. After seasonal differencing, spikes remain at lag 1. Propose full SARIMA model. 3) Compare ARIMA(1,1,1) and ARIMA(0,2,1) using AIC=120 vs 118. Which do you choose and why? 4) Residuals pass ACF visual check but Ljung-Box $p=0.03$. Interpretation? 5) Explain why over-differencing can harm forecast accuracy. Solutions included in appendix section of PDF.